

Adjustable Robust Optimization for Large-Scale Photovoltaics Planning in Smart Distribution Networks

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Abstract—High penetration in distributed photovoltaics (PVs) enhances the resilience of distribution network operations. However, it brings many uncertain issues such as the voltage and harmonic fluctuations that may affect the distribution network stability. Therefore, the determination of the PV location and capacity is of vital importance for smart distribution network planning. Considering the fluctuation of PV outputs and the robustness of optimization results, this paper proposes an adjustable robust optimization method for large-scale PV planning in smart distribution networks. The AC power flow is first converted into a convex program by the linearization theory, and the bi-level optimization is converted to a single-layer optimization problem using the strong duality theory. A hybrid algorithm based on the integration of grey wolf optimization (GWO) and improved particle swarm optimization (IPSO), named GWO-IPSO, is proposed to determine the robust optimal location and the capacity of distributed PVs. The proposed method is verified based on the IEEE 69-bus radial distribution network case study. The proposed method efficiently adapts to parameter uncertainties resulting from extreme scenarios and delivers better performance than general heuristic methods.

Index Terms—Distributed photovoltaic planning, adjustable robust optimization, bi-level optimization conversion, hybrid optimization algorithm.

I. INTRODUCTION

A. Background

THE urbanization in recent years has increased the energy demand for uninterrupted power supply to both domestic households and industries around the world. As a vital renewable source, solar energy has prominent development potential [1]. Photovoltaics (PV) systems play a significant role in the energy

conversion process when solar energy is harvested via solar cells [2]. Integrating large-scale PV into the smart distribution network (SDN) is an effective way to utilize local renewable resources, solve power shortages and reduce environmental pollution.

B. Motivation

The large-scale integration of distributed PVs is scattered in nature, and the variability and intermittency of the PV outputs may increase the risk of voltage violation. Appropriate placement of the PVs in a suitable capacity is pivotal for improved power quality for increasing the economic benefits of the SDN. Due to the dynamic nature of the distributed PV environment, the choice and placement of the PVs require extensive optimization [3]. Ehsan A et al. presented a comprehensive critical analysis of the advanced analytical approaches for optimal planning [4]. A bio-geography-based algorithm is proposed in [5] for optimal planning of multiple DGs to reduce the system network loss to reflect on the improvements in the system voltage profiles of the distribution network. Ref. [6] uses loss sensitivity factors to determine optimal DG capacities and locations for PVs. Ant lion optimization is adopted to adjust the DG capacities for improving voltage quality and reducing thermal loss. To minimize the costs of investment, maintenance, and CO₂ emission, a capacity optimization of turbines and PV power stations is presented in [7], where a mixed-integer second-order cone program (MISOCP) was formed based on the conic relaxation of power flow equations. However, the above studies ignore the uncertainty of PV output when planning PV capacity, which made it challenging to cope with the worst natural conditions.

C. Research Gap

The efficient mitigation of the dynamism of solar radiation should be considered during the initial PV planning. In the existing studies, fuzzy optimization, scenario-based stochastic optimization, and robust optimization are three primary techniques to handle PV uncertainties as follows:

- 1) The fuzzy optimization employs a fuzzy set to present constraints. The degree of satisfaction is defined as a membership function, whereas the optimization results are strongly influenced by the subjective arbitrariness of satisfaction degree [8], [9].

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- 2) To establish the system uncertainty model completely, the scenario-based stochastic optimization methods require statistics, analysis, and fitting of the measured data to calculate the probability distribution function (PDF) that can generate multiple scenarios [10]. Hakimi, SM et al. [11] propose a stochastic planning method to plan a multi-microgrid (MMG) operation in an electricity market considering the stochastic renewable energy resources (RERs) integration, resulting in the above 8% cost reduction in the MMG. In [12], a stochastic optimization method based on the Harris hawks optimization (HHO) algorithm is adopted to determine PVs' optimum capacity, number, and site to reduce real power losses and voltage deviation. However, the stochastic optimization methods rely heavily on the PDF of uncertain parameters. The precision required for PVs under dynamic scenarios often renders it difficult to converge for iterative algorithms. Furthermore, the optimal solution is often acquired for only a given PV scenario, which causes poor adaptability in various conditions [13]. Generally, the stochastic method requires statistics and analysis of historical data, and the enormous computational burden is un conducive to the rapid solution of the model [14].
- 3) Although the former two methods can provide more information than robust optimization, PDF is often challenging to obtain and calibrate in real distribution networks. For robust optimization methods, the uncertainty set can effectively reflect the uncertainties in the forecasting results, which can be readily determined by the existing forecasting methods. The existing prediction methods include random forest (RF) [15], time series [16], support vector machine [17], and neural networks [18], etc. Given the uncertainty through multiple deterministic bounded intervals, the robust optimization method has received more attention for generating an optimal solution that is minimally sensitive to uncertain factors [19]. Compared with the fuzzy and stochastic optimization methods, the optimal solution of robust optimization has low sensitivity to parameter changes, and the uncertain variables can be constructed by the uncertainty sets without the knowledge of the PDF. Because robust optimization does not need to generate many scenarios, it has a much lighter computational burden than the stochastic programming approach. Besides, the solution can satisfy the constraints under all scenarios and provide robustness against the worst case in the uncertainty set. Therefore, M. Aghamohamadi et al. [20] propose a robust co-optimization for the capacity allocation and bidding strategy of a prosumer equipped with PV, wind turbine, and battery energy storage. Block coordinate descent strategy is adopted to handle the inner max-min programming utilizing the Taylor series instead of transforming it into single-level max programming by duality theory. Considering the uncertainties of PV, T. Xu et al. [21] present a multi-objective robust optimization method for active distribution networks to maximize the utilization of renewables and minimize power losses. The non-cooperative and cooperative sequential games dominated by wind-PV cooperative alliance are proposed to

solve the optimal PV capacity allocations in [22]. The uncertainty set in the above robust optimization model allows all uncertain parameters to reach maximum or minimum values simultaneously. However, the situation is almost impossible in the real distribution network. It is necessary to make a moderate compromise on the uncertainty set of the robust optimization model to adjust the planning economics. Robust optimization in transmission network expansion planning [23] and optimal scheduling of power systems [24] have been widely used. Jabr, R. A presents a robust optimization approach for transmission network expansion planning under uncertainties of renewable generation and load [23]. Bertsimas, D et al. propose a two-stage adaptive robust unit commitment model for the security-constrained unit commitment problem in the presence of nodal net injection uncertainty [24]. However, these applications are based on DC power flow calculation methods. If these methods are applied to the SDN, it will be challenging to meet the many assumptions of DC power flow calculation on line impedance, node voltage, and branch phase angle. As a result, the calculation accuracy is significantly reduced and even does not meet the actual requirements. Therefore, it is necessary to study the robust optimal planning of SDN using the AC power flow model.

After establishing the model, solving the robust optimization model is also a challenge. The reasonable collocation between the optimization method and robust model is the key factor affecting the planning effectiveness. Most of the existing optimization methods are not suitable for robust optimization due to the non-convex and nonlinear relationships of SDN. In fact, the planning models can be formulated as constrained optimization problems comprising continuous and integer decision variables [25]. The derivative-based optimization algorithms may suffer from complex derivative computations and get stuck in local minimums [26]. Therefore, heuristic algorithms are generally preferred for calculating optimization problems because they provide a fast solution and are easy to understand and implement [26]. In [27], the moth swarm algorithm (MSA), biogeography-based optimization (BBO), and particle swarm optimization (PSO) are used to minimize the generating cost, respectively. Zhang, YL et al. [28] proposed an ant colony optimization method for power line carriers with the logistic function. Taha, IBM et al. [29] presented a new grey wolf optimizer (GWO) for sizing the optimal reactive power resources to enhance the power system operation.

The improvement of global search capabilities and accuracy remains the focus of heuristic algorithms, which is still drawing the attention of various researchers. However, there are still many gaps in searching for the optimal Pareto front of the problem and achieving all PV benefits with proper objective functions. The No-Free-Lunch (NFL) theorem states that, on average, the total possible benefits are the same for all algorithms for any optimization problem [30]. All meta-heuristic algorithms in the available studies found the near-optimal Pareto fronts [30]. However, they cannot guarantee the global minimum because the algorithms run for a finite number of runs and iterations. Besides, The NFL theorem states that a single heuristic algorithm cannot solve all possible optimization problems. Therefore, heuristic

algorithms must be tuned and implemented for specific problems [31]. From the available studies that aimed to improve the GWO, easy fall into the local optimum points and weak exploration phases are still key challenges because search agents of GWO always update their positions based on only the top three fittest members of the group [32]. The other members do not play a substantial role in generating a new solution and thus making the algorithm stuck in local optimum points [32]. It is necessary to combine other technologies to improve the situation.

D. Contribution

In this paper, a PV adjustable robust planning model is proposed, which focuses on the processing of solar radiation uncertainty, PV sizing, and PV placement in a distribution network. Using the inverse function of solar radiation cumulative distribution, the adjustable robust optimization model converts the uncertain variables into interval sets. According to the linearization theory, the economic objective function and constraints are transformed into a robust bi-level optimization problem. Then, the bi-level optimization problem can be converted to an equivalent single-layer optimization problem based on the strong duality theory. Besides, a hybrid algorithm combining the grey wolf optimization and an improved particle swarm optimization (GWO-IPSO) is presented to solve the robust optimization problem. Eventually, the practicability of the proposed model is verified through the IEEE 69-bus radial distribution network. The significant contributions of this paper are listed as follows:

- 1) Considering the uncertainty of solar radiation, an adjustable robust optimization is proposed for large-scale PV planning in SDNs. The proposed optimization balances the trade-off between the risks involved and the maximum level of conservative solution, using a computationally tractable strategy. The economics of SDN operation can be guaranteed in normal or extreme conditions with the proposed approach, which improves the disadvantage of existing stochastic and deterministic planning that is not robust and economical enough in the case of insufficient light.
- 2) This paper guarantees the robustness and feasibility of PV planning by defining reasonable PV uncertainty set, economic objective functions, and linearized constraints, which renders it unnecessary to obtain the PDF of distributed PV outputs. Due to the low accuracy of the common DC power flow model in distribution networks, the linearized AC power flow model is introduced for robust optimization of distribution networks. Besides, the linearization theory and strong duality theory are employed to deal with robust bi-level optimization problems to reduce the solving complexity and improve optimization efficiency.
- 3) This paper proposes a novel integration of the grey wolf optimization with an improved particle swarm optimization (GWO-IPSO), which efficiently deals with the premature convergence problem of PSO, mitigates the tendency of the GWO to fall into the local optimum, and eliminates

the dependence of robust optimization models on the nonlinear solver.

E. Organization

The rest of the paper is organized as follows. First, the overall distributed PV planning framework is described in Section II. Section III introduces the proposed adjustable PV robust optimization model. The proposed optimization approach is introduced in Section IV, and Section V verifies our method based on an IEEE 69-bus radial distribution network. Section VI concludes this paper.

II. DISTRIBUTED PV PLANNING FRAMEWORK

The distributed PV planning model involves objective function, constraints, robust optimization, solar radiation uncertainty, PV sizing, and placement. This paper proposes a general structure for PV planning, as shown in Fig. 1, which elaborates on the implementation process of robust planning. Compared with traditional planning methods, the proposed scheme can adapt to more dynamic scenarios and accordingly adjust the planning conservativeness. The intricate solution process of robust optimization is mathematically simplified in the proposed GWO-IPSO method, which further improves the economic benefits. The objective function and constraints of linearization also guarantee the practicality of planning.

A. Decision Variables

The robust planning model requires the power injected by distributed PVs, which depends on solar radiation and PV capacities [33]. Statistical models and artificial intelligence technology are broadly applied in solar radiation forecasting [34], [35]. It is worth mentioning that the distributed PV outputs are almost certain if considering only the PV capacity but ignoring the solar radiation fluctuation. To eliminate unknown variables during the PV planning, we consider additional decision variables, including distributed PV allocations L_{i-PV} and capacities C_{i-PV} .

B. Objective Function

The objective function includes active power loss, distributed PV investment, and distributed PV operating cost, which is defined in (1).

$$\begin{aligned} \min f = & C_{\text{loss}} \sum_{i=1}^j \sum_{j \in i} \text{Real} \left| (U_i - U_j)^2 / R_{ij} - jX_{ij} \right| \\ & + C_{\text{PV-sum}} [(r_d(1+r_d)^n) / ((1+r_d)^n - 1) C_{\text{in}} + C_{\text{op}}] \end{aligned} \quad (1)$$

where C_{loss} is the cost of unit power loss; $j \in i$ represents all the nodes that are connected to node i . U_i is the voltage of node i ; $\text{Real}()$ depicts consideration of real parameters; R_{ij} and X_{ij} represent the resistance and reactance of the line between buses i and j , respectively; $C_{\text{PV-sum}}$ is the total distributed PV capacity connected to the distribution network; r_d is the discount rate; n is the service life of distributed PVs, and C_{in} and C_{op}

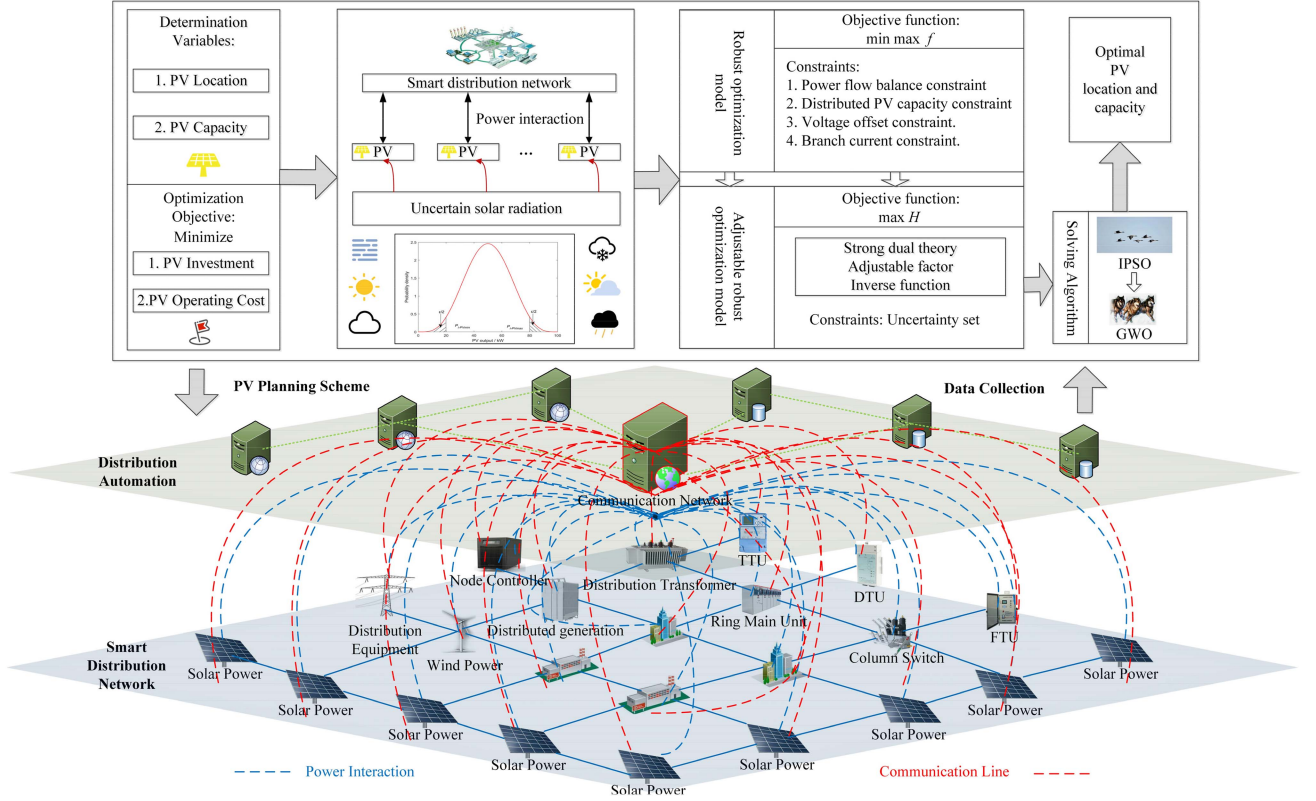


Fig. 1. The general structure of distributed PV planning model.

are the investment and operating costs of the unit PV capacity, respectively.

C. Constraints

Constraint 1: Power flow balance.

The distributed PV power generation and the injected power from the grid should meet the power flow constraints of the power distribution system [36], i.e.,:

$$\begin{cases} \sum_{i=1}^N P_{i-PV} + P_{grid} = P_L + P_{Loss} \\ \sum_{i=1}^N Q_{i-PV} + Q_{grid} = Q_L + Q_{Loss} \end{cases} \quad (2)$$

where Q_{i-PV} and P_{i-PV} are the reactive and active power injected by the distributed PV connected to node i ; Q_{grid} and P_{grid} are the reactive and active power supplied by the grid, respectively; Q_L and P_L are the reactive and active power required by the load. Q_{Loss} and P_{Loss} are the reactive and active power losses.

Constraint 2: Distributed PV capacity.

The capacity of distributed PVs is limited by multiple factors. A high-distributed PV penetration rate reduces the stability of the distribution network. Consequently, the capacity of the distributed PVs connected to the distribution network should be limited, which can be expressed as follows:

$$\sum_{n=1}^N C_{i-PV} \leq C_{PVmax} \quad (3)$$

$$0 \leq C_{i-PV} \leq C_{i-PVmax} \quad (4)$$

where C_{i-PV} and $C_{i-PVmax}$ are the current and maximum distributed PV capacity of node i based on local PV policy; C_{PVmax} is the maximum distributed PV capacity in the distribution network. $C_{i-PVmax}$ is set according to the maximum reverse power of the branch and transformer. C_{PVmax} can be used to limit the distributed PV penetration rate of the distribution network, which is determined by the local operational management strategy and planning strategy.

Constraint 3: Voltage offset constraint.

$$93\% |U_i| \leq |U_i| \leq 107\% |U_i|, i = 1, \dots, N \quad (5)$$

where U_i is the voltage of node i .

Constraint 4: Branch current.

$$|I_l| \leq I_{l-max} \quad l = 1, 2, \dots, N_l \quad (6)$$

where I_l is the current of branch l ; N_l is the branch number in the distribution network, and I_{l-max} is the current upper limits of branch l .

Constraint 5: Power flow equation.

$$P_{ij} + jQ_{ij} = (G_{ij} - jB_{ij}) (|U_i|^2 - U_i U_j^*) \quad \forall ij \in \Omega_l \quad (7)$$

$$I_{ij}^2 = (G_{ij}^2 + B_{ij}^2) (|U_i|^2 + |U_j|^2 - 2\text{Real}\{U_i U_j^*\}), \quad \forall ij \in \Omega_L, \forall i \in \Omega_N \quad (8)$$

where Q_{ij} and P_{ij} are the reactive and active power of the branch between nodes i and j ; Ω_l is the line set of the distribution

network; B_{ij} and G_{ij} are the susceptance and conductance of the branch between nodes i and j .

D. Linearization of Power Flow

As mentioned in the Introduction, DC power flow calculation is no longer applicable to distribution networks with many branches and nodes. As conventional AC power flow calculation is a non-convex nonlinear problem, it is challenging to solve the AC power flow calculation considering its robustness. Although the heuristic optimization algorithms can solve it directly, their efficiency and accuracy are challenging to guarantee. To transform the AC power flow into a convex program, this paper adopts the conic power flow formulation to equivalent AC power flow calculation [37], then approximates the conic constraints to a linear problem by a polyhedral approximation method [38], and formulates application conditions for strong duality theory [39] in Section III.

A Hermitian matrix M satisfying $M = M^*$ is constructed, M_{ij} represents the element in the i th row and j th column of M .

$$M = \begin{bmatrix} |U_1|^2 & \cdots & U_1 U_i^* & \cdots & U_1 U_n^* \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ U_i U_1^* & \cdots & |U_i|^2 & \cdots & U_i U_n^* \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ U_n U_1^* & \cdots & U_n U_i^* & \cdots & |U_n|^2 \end{bmatrix} \quad (9)$$

Then, (5), (7), and (8) can be rewritten as the line power flow (10), the current magnitude squared (11), and the voltage magnitude squared (12).

$$\begin{cases} P_{ij} = G_{ij} (M_{ii} - \text{Real}\{M_{ij}\}) - B_{ij} \text{IM}\{M_{ij}\} \\ Q_{ij} = -B_{ij} (M_{ii} - \text{Real}\{M_{ij}\}) - G_{ij} \text{IM}\{M_{ij}\} \end{cases} \quad (10)$$

$$I_{ij}^2 = (G_{ij}^2 + B_{ij}^2) (M_{ii} + M_{jj} - 2\text{Real}\{M_{ij}\}) \quad (11)$$

$$I_l^2 \leq I_{l-\max}^2 \quad (12)$$

$$(U_i^{\min})^2 \leq M_{ii} \leq (U_i^{\max})^2 \quad (13)$$

where $\text{IM}()$ denotes the imaginary parameters. If the matrix is positive semi-definite and load over-satisfaction are allowed, the presented model given by minimizing (1) subject to constraints (1)–(8) is equivalent to the original problem of minimizing (1) subject to (2)–(4), (10)–(13). The semi-definite constraint in the presented model can be further replaced with a set of rotated conic quadratic constraints, which can be defined as:

$$\begin{cases} M_{ii} M_{jj} \geq \text{Real}\{M_{ij}\}^2 + \text{IM}\{M_{ij}\}^2 \\ \text{Real}\{M_{ij}\} = \text{Real}\{M_{ji}\} \\ \text{IM}\{M_{ij}\} = -\text{IM}\{M_{ji}\} \end{cases} \quad (14)$$

Each inequality in (14) can be expressed as:

$$r_1 r_2 \geq x_0^2 + y_0^2, \quad r_1, r_2 \geq 0 \quad (15)$$

To achieve linearization, the rotation cone constraint split is divided into the following two cone constraints and three

equations as follows:

$$\begin{cases} r \geq \sqrt{x_0^2 + y_0^2} \\ r' \geq \sqrt{(x'_0)^2 + (y'_0)^2} \\ r' = (r_1 + r_2)/2 \\ x'_0 = (r_1 - r_2)/2 \\ y'_0 = r \end{cases} \quad (16)$$

We introduce an integer parameter $\omega > 2$ and 2ω variables ($x_i, i = 1, \dots, \omega$ and $y_i, i = 1, \dots, \omega$), and a polyhedron can be constructed as follows:

$$x_{i+1} - x_i \cos \frac{\pi}{2\omega} - y_i \sin \frac{\pi}{2\omega} = 0, \quad i = 0, \dots, \omega - 1 \quad (17)$$

$$y_{i+1} - y_i \cos \frac{\pi}{2\omega} + x_i \sin \frac{\pi}{2\omega} \geq 0, \quad i = 0, \dots, \omega - 1 \quad (18)$$

$$y_{i+1} + y_i \cos \frac{\pi}{2\omega} - x_i \sin \frac{\pi}{2\omega} \geq 0, \quad i = 0, \dots, \omega - 1 \quad (19)$$

$$x_\omega \cos \frac{\pi}{2\omega} + y_\omega \sin \frac{\pi}{2\omega} - r = 0 \quad (20)$$

F. Glineur et al. [38] demonstrate that the equivalent accuracy of the above-mentioned polyhedral approximation cone in (16) is $\cos(\pi/2^\omega)^{-1} - 1$.

III. ADJUSTABLE ROBUST OPTIMIZATION MODEL

Wen Y et al. present two probabilistic approaches to estimate the uncertainty associated with PV power point forecasts. By using the radial basis function neural network (RBFNN) [40], the solar radiation and distributed PV outputs in Section II can be determined. However, the actual PV outputs are often intermittent and variable. Herein, the operating effectiveness of the planning scheme can be made consistent with the actual situation by exploiting historical insights and future load forecasts. To improve the robustness, the uncertainty of solar radiation should be considered when optimizing the distributed PV location and capacity during the planning. To this end, this paper presents an adjustable robust optimization model.

Robust optimization quantifies the variable uncertainty in a set, and any decision variable that takes a value in the set can make optimization results subject to all constraints. To find the optimal decision variables in the worst scenarios, the adjustable robust optimization model should be constructed based on the worst-case scenarios. In this paper, the economic objective function and constraints in Section II are converted to a robust programming model, which can be further transformed into an equivalent single-layer optimization problem with the strong duality theory. Finally, the GWO-IPSO method is then adopted to solve the optimization problem.

A. Uncertainty Set

The output power of PV is determined by the PV capacity and solar radiation at different periods. The solar radiation u approximately follows the β -distribution [41], from which the probability density of the PV output is obtained as follows:

$$f(P_{i-PV}) = B(a_t, b_t)^{-1} (P_{i-PV}/C_{i-PV})^{a_t-1} (1 - P_{i-PV}/C_{i-PV})^{b_t-1} \quad (21)$$

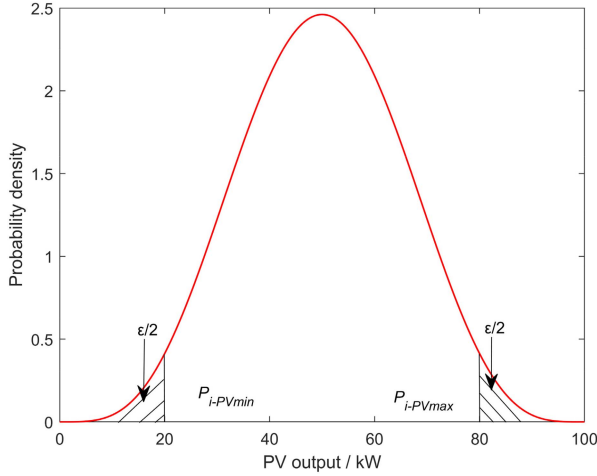


Fig. 2. The PDF of PV outputs.

$$\begin{cases} a_t = \mu_{i-PV} [\mu_{i-PV} (1 - \mu_{i-PV}) / \sigma_{i-PV}^2 - 1] \\ b_t = (1 - \mu_{i-PV}) [\mu_{i-PV} (1 - \mu_{i-PV}) / \sigma_{i-PV}^2 - 1] \end{cases} \quad (22)$$

where σ_{i-PV} and μ_{i-PV} are the standard deviation and mean of the PV output power, respectively; $B(a_t, b_t)$ represents the Beta Function. Similarly, for a given confidence $1-\varepsilon$, the inverse function of the PV output cumulative distribution function can be employed to determine the uncertainty set of PV output. As shown in Fig. 2, each solar radiation scenario corresponds to a specific ratio of PV output power. $P_{i-PVmin}$ and $P_{i-PVmax}$ are the lower and upper limits of the PV outputs.

Using the given forecasting method, the estimated and fluctuating value of the solar radiation (u_{ave} and ξ_u) can be determined. The variation interval of the solar radiation is $\{u | u_{ave} - \xi_u \leq u \leq u_{ave} + \xi_u\}$. According to the given confidence $1-\varepsilon$, we can obtain $P_{i-PV} \in [P_{i-PVmin}, P_{i-PVmax}]$ by using the inverse function of solar radiation cumulative distribution. For the robust optimization model, u is the uncertain parameter. u_{min} and u_{max} are the lower and upper limits of solar radiation, respectively, and U represents the uncertainty set.

B. Adjustable Robust Optimization

The bi-level optimization problem is a particular optimization problem with hierarchical characteristics, which is challenging to solve with heuristic algorithms. Compared to directly solving the bilevel optimization problem, the advantages of converting it into a single-level optimization problem and then solving it using heuristic algorithms are: (1) Converting the bilevel optimization problem into a single-level optimization problem can greatly simplify the problem, reducing the algorithmic and computational complexity. Equation (2) Directly solving the bilevel optimization problem requires iterative calculations, while converting it into a single-level optimization problem can avoid a large number of iteration calculations, thereby reducing the computation. Equation (3) Heuristic algorithms for solving single-level optimization problems are usually more efficient than directly solving bilevel optimization problems. Heuristic algorithms can improve solution efficiency through parallel

computing and optimization of search algorithms, whereas the direct solution of bilevel optimization problems may require more time and computational resources.

The upper level and the lower level of the bi-level optimization problem influence and restrict each other. The upper level optimizes the objective function independently and is bound by the lower-level results [42]. Robust optimization is a bi-level optimization model [43], whose simplified expression is as follows:

$$\text{Min}_x \left[\text{Max}_u c^T x \right] \quad (23)$$

Subject to

$$H(u)x = h, \quad \forall u \in U \quad (24)$$

$$Gx \leq g \quad (25)$$

$$x_{inf} \leq x \leq x_{sup} \quad (26)$$

where $H(u)$ is a coefficient matrix of u ; x is the decision variable; c is the coefficient matrix of the objective function; (25) represents the equality constraints, (26) represents the inequality constraints, and represents the upper and lower bound of the decision variable. In a large-scale SDN, the number of decision variables x is significantly higher than that of uncertain variable u , resulting in unsatisfactory optimization efficiency. To improve optimization convergence, the optimization sequence in the objective function (23) can be exchanged. According to [44], the objective function (23) is equivalent to (27):

$$\text{Max}_u \left[\text{Min}_x c^T x \right] \quad (27)$$

The two objective functions contradict each other, which causes difficulty in the solving process. According to the linearization theory and strong duality theory [39], (24)–(27) can be further transformed into (28)–(29).

$$\text{Max}_u \left[\text{Max}_{\alpha, \beta, \gamma_1, \gamma_2} h^T \alpha + g^T \beta + x_{inf}^T \gamma_1 + x_{sup}^T \gamma_2 \right] \quad (28)$$

Subject to

$$H(u)^T \alpha + G^T \beta + \gamma_1 + \gamma_2 = 0 \quad (29)$$

where $\alpha, \beta, \gamma_1, \gamma_2$ are dual variables, α is a free variable, $\beta \leq 0$, $\gamma_1 \geq 0$, and $\gamma_2 \leq 0$. The constraints can be substituted into the objective function to obtain (30):

$$\text{Max}_u \left\{ \text{Max}_{\alpha, \beta, \gamma_2} \left\{ \left[h^T - x_{inf}^T H(u)^T \right] \alpha + (g^T - x_{inf}^T G^T) \beta + (x_{sup}^T - x_{inf}^T) \gamma_2 \right\} \right\} \quad (30)$$

Therefore, the bi-level maximum optimization can be converted to a single-layer optimization problem as follows:

$$\text{Max}_{u, \alpha, \beta, \gamma_2} \left\{ \left[h^T - x_{inf}^T H(u)^T \right] \alpha + (g^T - x_{inf}^T G^T) \beta + (x_{sup}^T - x_{inf}^T) \gamma_2 \right\} \quad (31)$$

Subject to

$$u \in U = \{u_{min}, u_{max}\}, \beta \leq 0, \gamma_1 \geq 0, \gamma_2 \leq 0 \quad (32)$$

where $H(u)^T \alpha$ is a bilinear function. When u takes the extreme value of the set U , $H(u)^T \alpha$ reaches the extreme value.

The situation where the uncertain parameters reach the worst value simultaneously is a small probability event, making the optimization results deviate from the actual situation. To overcome the shortcoming, this paper makes a moderate compromise on the uncertainty set and realizes adjustable robust optimization. The uncertainty set can be converted according to (33):

$$U = \left\{ u \mid u_i = u_{ave-i} + \xi_{u-i} \mu_i, \sum_{i=1}^{N_{un}} |\mu_i| \leq \Gamma, |\mu_i| \leq 1 \right\} \quad (33)$$

where μ_i is the adjustable factor (0 or 1) of the i th uncertain variable, and Γ is the overall adjustable margin. A reasonable value of adjustable margin Γ can limit the system uncertainty to a normal level. Equation (33) is substituted into (31) to obtain the following adjustable robust optimization model:

$$\begin{aligned} \max_{\alpha, \beta, \gamma_2} & \left[h^T \alpha - x_{inf}^T u_{sup}^T \alpha + \min \sum_{i=1}^{N_{un}} \xi_{u-i} \alpha_i \mu_i x_{inf}^T \right. \\ & \left. + (g^T - x_{inf}^T G^T) \beta + (x_{sup}^T - x_{inf}^T) \gamma_2 \right] \end{aligned} \quad (34)$$

where N_{un} is the number of random variables.

Assuming that α_k , β_k , and $\gamma_{2,k}$ are α , β , and γ_2 at the k th iteration, respectively. x_k and u_k are decision and uncertain variables at the k th iteration, respectively. Based on the above process, the flow chart of the proposed robust optimization model is shown in Fig. 3.

IV. HYBRID ALGORITHM OF PARTICLE SWARM OPTIMIZATION AND IMPROVED GREY WOLF OPTIMIZATION

Because the objective function in Section III contains a bilinear term, the conventional Mixed Integer Linear Programming (MILP) solver cannot directly calculate the objective function. Although convex relaxation can relax a Mixed Integer Non-Linear Programming (MINLP) problem into an MNL problem, the optimal solution in the relaxed problem is not always the optimal solution to the objective function [45]. It is challenging to determine a convex relaxation with the tightest bounds. The heuristic optimization algorithms are independent of convex relaxation and alleviate this phenomenon.

Among the popular heuristic methods, PSO is a robust evolutionary computation algorithm that works based on the cooperation and experience of swarms, which has easy implementation and satisfactory robustness to control parameters and computational efficiency. Although it has numerous advantages, it gets trapped in a local minimum and leads to premature convergence when solving heavily constrained problems due to the limited local and global searching capabilities [46], [47]. To deal with this condition, hybrid PSO is drawing increasing attention [48]. Motivated by this observation, this paper introduces variable inertia weights into the traditional PSO and constructs the IPSO, which can improve the convergence accuracy of heuristic optimization algorithms. To further address the premature convergence problem of PSO, GWO is integrated into IPSO to develop the proposed GWO-IPSO method due to its positive balance between exploitation and exploration. Meanwhile, the randomness and dispersion of the particle swarm movement of the IPSO can

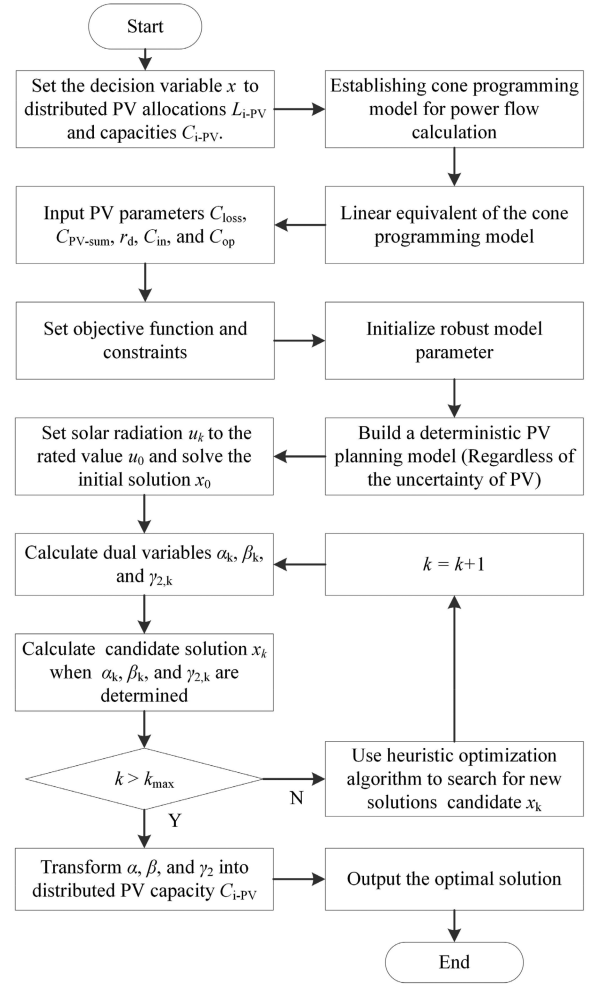


Fig. 3. Flow chart of the robust optimization model.

mitigate the tendency of the GWO to fall into the local optimum. In other words, GWO is used to find the global optimal solution, and then IPSO is used to further optimize the local optimal solution to obtain a better solution. Therefore, GWO and IPSO can complement each other and achieve better performance in solving complex optimization problems. This GWO-IPSO combination algorithm can fully utilize their respective strengths and stabilizes robust optimization effectiveness. The process flow of the GWO-IPSO algorithm is shown in Fig. 4, and the calculation steps are as follows:

Step 1: Initialize the particle swarm.

The input variables are designated as α , β , and γ_2 proposed in Section III, which are used for the solution candidates. Specifically, α is a free variable, $\beta \leq 0$, and $\gamma_2 \leq 0$. The position of the particle is set as a vector of candidate solutions (including α , β , and γ_2). The original upper and lower bounds of the C_{i-PV} are incorporated into the objective function, which is stated in (4). According to the particle's upper and lower bounds, calibration metrics including particle position, velocity, personal best position P_{best} , global best position G_{best} , and initial fitness can be initialized randomly. P_{best} is the position of the highest fitness returned for a specific particle (including α , β , and γ_2),

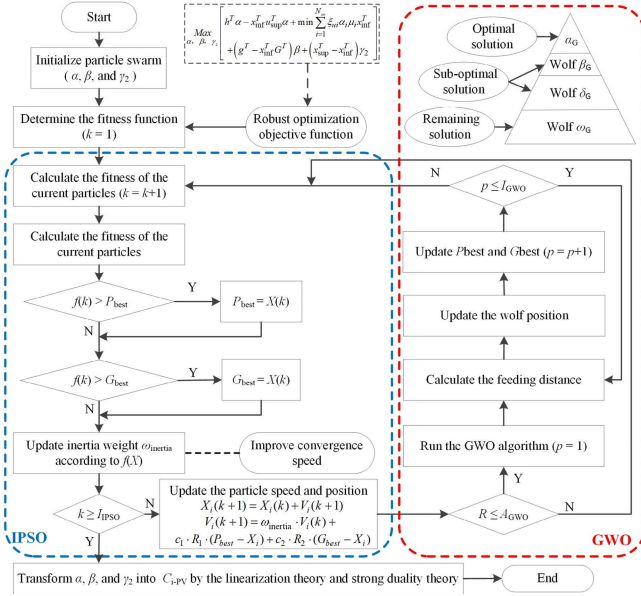


Fig. 4. Flow chart of GWO-IPSO.

and G_{best} is the position of the highest fitness returned for the whole swarm (including α , β , and γ_2).

Step 2: Determine the fitness function.

The objective function (34) presented in Section II is determined as the fitness function of the iteration. The smaller the value of the fitness function, the better. k represents the iteration number, and now the first IPSO iteration ($k = 1$) is performed.

Step 3: Calculate the fitness of the current particles.

All the particles follow the same principle: while continuously acquiring the fitness f (the effectiveness of the given solution) of their current position, particles accelerate toward P_{best} and G_{best} .

Step 4: Update P_{best} and G_{best} .

Suppose a particle's fitness $f(k)$ at its current position is less than the fitness value at its respective P_{best} or G_{best} . The current position is replaced with P_{best} or G_{best} accordingly, whichever is lower.

Step 5: Update inertia weight $\omega_{inertia}$ according to $f(k)$.

In this step, the inertia weight of PSO is improved. According to the optimization demand for different particle fitness, the variable inertia weight is established. While dynamically regulating $\omega_{inertia}$, fine particles tend to perform exploitation to refine the local search results, and weak particles perform extensive modifications to explore space in massive steps. Therefore, the probability of falling into a local optimum can be reduced. f_c represents the average fitness of the current particle. f_p represents the average fitness of the global best particle.

1) When $f(k) < f_c$, the current particle is far from G_{best} , and the inertia weight should be larger. $\omega_{inertia}$ is calculated as follows:

$$\omega_{inertia} = \omega_{avg} + rand(0, 1) \times (\omega_{max} - \omega_{min})/2 \quad (35)$$

$$\omega_{avg} = (\omega_{max} + \omega_{min})/2 \quad (36)$$

where ω_{min} and ω_{max} are the minimum and maximum inertia weight, respectively; ω_{avg} is the average inertia weight.

2) When $f(k) > f_p$, the current particle is closer to G_{best} , and the inertia weight should be smaller. $\omega_{inertia}$ is calculated as follows:

$$\omega_{inertia} = \omega_{avg} - |[f(k) - f_p]/(P_g - f_p)| \times (\omega_{avg} - \omega_{min}) \quad (37)$$

When $f_c < f(k) < f_p$, a nonlinear decreasing trend of inertia weight is adopted as follows:

$$\omega_{inertia} = \omega_{max} - k(\omega_{max} - \omega_{min})/I_{IPSO} \quad (38)$$

where I_{IPSO} is the maximum number of IPSO iterations.

Step 6: Update the iteration counter ($k = k + 1$).

If the IPSO iteration number k reaches I_{IPSO} , output G_{best} as the optimal solution, and transform α , β , and γ_2 of G_{best} into distributed PV capacity C_{i-PV} by the linearization theory and strong duality theory [39].

Step 7: Update the particle position and velocity.

The updated formulas of the particle's position can be expressed as:

$$X_i(k+1) = X_i(k) + V_i(k+1) \quad (39)$$

where $X_i(k)$ stands for the position of particle i in the k th IPSO iteration. $V_i(k)$ is the velocity of particle i in the k th IPSO iteration, which is the key factor of the optimization. The particle velocity is accelerated based on the relative positions of P_{best} and G_{best} . The goal of $V_i(k+1)$ is to guide the particles to move to the position with the best fitness, which is defined as:

$$V_i(k+1) = \left[\omega_{inertia} \cdot V_i(k) + c_1 \cdot R_1 \cdot (P_{best} - X_i) + c_2 \cdot R_2 \cdot (G_{best} - X_i) \right] \quad (40)$$

where R_1 and R_2 are the random numbers in the range of $[-1, 1]$; c_1 and c_2 are the self-learning factors.

Step 8: If $R \leq A_{GWO}$, go to step 9; Otherwise, go to Step 3.

R is a random coefficient in $[0, 1]$. A_{GWO} is the GWO startup parameter (valued from 0 to 1), which can be adjusted according to the actual running time and effectiveness. When A_{GWO} is 1, P_{best} particles will be transformed into grey wolves (including α , β , and γ_2) for global search at every IPSO iteration.

Step 9: To assist the particle swarm to jump out of the local optimum, perform the GWO algorithm only on the P_{best} of IPSO.

GWO is a heuristic search method inspired by grey wolf behaviors [49]. To deal with the non-convex optimization problem, the stratum and hunting nature of the grey wolf is adopted. GWO mimics the hunting behavior of the wolves, which includes three wolves α_G , β_G , δ_G , and the remaining wolves ω_G . The position of each wolf consists of α , β , and γ_2 . The primary process involves searching for the prey, surrounding the prey, and attacking the prey. α_G represents the optimal solution; β_G and δ_G represent the sub-optimal solutions. ω_G represents the remaining candidate solutions. The wolf pack starts from the initial solutions of the three wolves α_G , β_G , and δ_G , and gradually approaches the optimal solution of the

objective function in the solution space. Positions of the wolves are then updated and corrected while gradually shrinking to the vicinity of the prey until the optimal solution is reached. To reduce the computational complexity, GWO is only used to optimize the P_{best} of IPSO to enhance its local searchability. A particle swarm composed of P_{best} will be transformed into a grey wolf population (including α , β , and γ_2), and the number of grey wolves N_{wolf} is initialized in this step.

Step 10: Calculate the feeding distance.

$D_{k \text{ id}}$ represents the feeding distance between the grey wolf and the prey at the p th GWO iteration, which is calculated as follows:

$$D_{\text{id}}^p = \left| C \times X_{\text{pd}}^p - X_{\text{id}}^p \right| \quad (41)$$

where $X_{\text{p id}}$ and $X_{\text{p pd}}$ are the locations of the grey wolf and the prey at the p th GWO iteration. Pongage coefficient C can be expressed as:

$$C = 2 \times r_3 \quad (42)$$

where r_3 is a random coefficient in $[0, 1]$.

Step 11: Update the wolf position.

As the distance $D_{\text{p id}}$ decreases, the individual wolf position is constantly updated by:

$$X_{\text{id}}^{p+1} = X_{\text{pd}}^p - A \times D_{\text{id}}^p \quad (43)$$

$$A = a \times (2r_4 - 1) \quad (44)$$

where the convergence factor A directly affects the global search capability of GWO, the coefficient a decreases linearly from 2 to 0 as the iteration number increases, and r_4 is another random coefficient in $[0, 1]$.

Step 12: Update the prey position.

The grey wolves (including α , β , and γ_2) search for the prey position under the leadership of the wolf pack. Positions of the remaining wolves ω_G are as follows:

$$\begin{cases} D_\alpha = |C_1 X_\alpha^p - X_{\text{id}}^p| \\ D_\beta = |C_2 X_\beta^p - X_{\text{id}}^p| \\ D_\delta = |C_3 X_\delta^p - X_{\text{id}}^p| \end{cases} \quad (45)$$

where $X_{\text{p } \alpha}$, $X_{\text{p } \beta}$, and $X_{\text{p } \delta}$ are the immediate positions of the three grey wolves α_G , β_G , and δ_G in the p th GWO iteration. The prey position $X_{\text{p+1 pd}}$ is calculated as:

$$\begin{cases} X_1^p = |X_\alpha^p - a_1 D_\alpha^p| \\ X_2^p = |X_\beta^p - a_2 D_\beta^p| \\ X_3^p = |X_\delta^p - a_3 D_\delta^p| \end{cases} \quad (46)$$

$$X_{\text{pd}}^{p+1} = (X_1^p + X_2^p + X_3^p) / 3 \quad (47)$$

where $D_{\text{p } \alpha}$, $D_{\text{p } \beta}$, and $D_{\text{p } \delta}$ are the distance between the head wolf in the remaining wolves ω_G and the individual wolf, respectively.

Step 13: Update P_{best} and G_{best} . if the fitness of $X_{\text{p+1 pd}}$ is greater than that of P_{best} , the current P_{best} of IPSO is replaced by the prey position $X_{\text{p+1 pd}}$.

Step 14: If the number of the GWO iterations p reaches the maximum number I_{GWO} , go to Step 3 and use the global

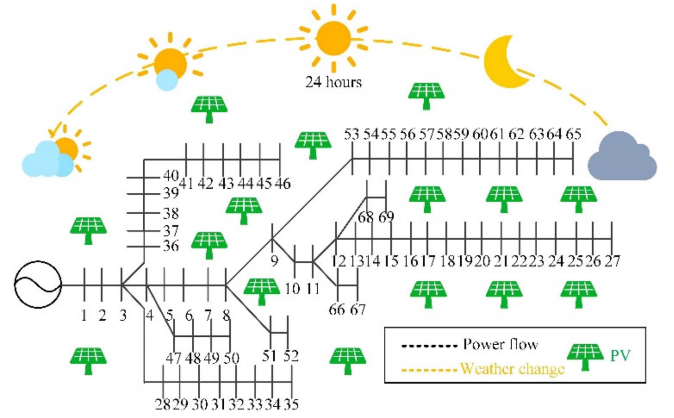


Fig. 5. 69-bus radial distribution network.

TABLE I
OPTIMIZATION PROBLEM PARAMETERS

Parameters	Value	Parameters	Value
C_{in}	1200 (\$/kW)	$C_{\text{i-PVmax}}$	40 (kW)
C_{loss}	1390 (\$/kW)	C_{PVmax}	1000 (kW)
n	20 (year)	N_{wolf}	30
r_d	0.1	N_{par}	30
c_2	1	I_{IPSO}	200
c_1	1	I_{GWO}	50
Γ	68	$C_{\text{i-PVmin}}$	0 (kW)

dispersion of particles to avoid GWO from falling into a local optimum; otherwise, go to Step 9.

V. CASE STUDIES

To verify the proposed method, an IEEE 69-bus radial distribution network, as shown in Fig. 5, is selected for PV robust planning [1]. The parameters of robust optimization are presented in Table I, which refers to the management policy of the State Grid of China in Binhai District, Tianjin.

A. Hyper-Parameter Optimization

The learning factors determine the direction of the particle search, which significantly affects the balance between exploration and exploitation. Therefore, the preferences of learning factors c_1 and c_2 are discussed in this Section. Increasing c_1 encourages exploration of the solution space as each particle moves toward the respective P_{best} ; increasing c_2 encourages exploitation of the supposed global optimum. Suppose that the solar radiation is 500 W/m^2 , the objective function value under different c_1 and c_2 is shown in Fig. 6. The smaller the objective function value, the better. Due to the randomness in the optimization process of the heuristic algorithm, the optimization effectiveness under different hyper-parameters fluctuates significantly. Through the experiments, we found that when $c_1 = 1$ and $c_2 = 0.4$, the optimization effectiveness is at its best. In the subsequent tests, this paper employs the above default hyperparameter settings.

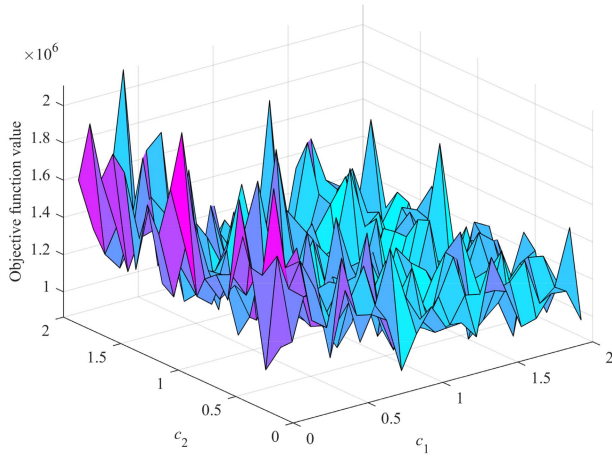


Fig. 6. The objective function value under different c_1 and c_2 .

B. Scenario Comparison

Most of the PV planning schemes currently applied use deterministic schemes that do not consider the uncertainty of solar radiation. To analyze the significance of PV robust planning for practical applications, this paper compares several uncertain schemes and one deterministic scheme. We simulate multiple uncertain scenarios of solar radiation, as listed below:

Scenario 1: Deterministic scenarios considered in the conventional scheme. When the fluctuation of solar radiation is ignored, u_{ave} and ξ_u are 521 W/m² and 0 W/m², respectively.

Scenario 2: Uncertain scenario. $u_{ave} = 482$ W/m², and $\xi_u = 53$ W/m².

Scenario 3: Uncertain scenario. $u_{ave} = 378$ W/m², and $\xi_u = 68$ W/m².

Scenario 4: Uncertain scenario. $u_{ave} = 521$ W/m², and $\xi_u = 73$ W/m².

The solar radiation interval of a scenario can be determined by the prediction methods. Regardless of the adjustability of robust optimization, PVs are planned to be installed at all nodes except node 1. When μ_i ($i = 1, 2, \dots, 68$) is 1 and Γ is 68, the robust planning results are shown in Fig. 7.

To analyze the difference between robust and deterministic optimization, the deterministic Scenario 1 and the uncertain Scenario 2-4 are compared. Scenario 1 is the conventional PV planning scenario, which does not consider the uncertainty of solar radiation. Compared with Scenario 1, the planning of the PV capacity of each node in Scenario 2-4 has changed due to the uncertainty of solar radiation, and the increase in the PV capacity of nodes 46-55 is conducive to improving the robustness of planning. From Scenario 2 and Scenario 3, as the environment becomes worse and the uncertainty interval increases, the PV capacity installed in the distribution network also increases. A harsh environment makes the installed PV insufficient to compensate for the nearby load, thus greater PV capacity would be required. Comparing the four scenarios, the PV installed on nodes 49-52 always has the largest capacity. Consequently, these three nodes are the key to determining the power quality and the operational robustness of the distribution network.

TABLE II
COMPARISON OF EXTREME AND NORMAL ENVIRONMENT

Method	Scenario for PV scheme formulation	Definite solar radiation (W/m ²)	Confidence coefficient	Objective function value (\$)
The proposed method	Scenario 1 (deterministic)	$u = 521$	/	1037162
		$u = 400$	/	1348968
		$u = 300$	/	2237795
	Scenario 2	$u = 482$	/	1134988
		$u = 400$	/	1276483
		$u = 300$	/	1852033
	Scenario 3	$u = 378$	/	1209182
		$u = 300$	/	1843233
		$u = 521$	/	1075793
	Scenario 4	$u = 400$	/	1234656
		$u = 300$	/	1951022
		$u = 521$	0.85	963278
Stochastic method	Scenario 4	$u = 400$	0.85	1293443
		$u = 300$	0.85	2192315

The performance of the four schemes in various environments (extreme and normal solar radiation) is shown in Table II. The PV planning scheme of Scenario 2-4 consider the uncertainty of solar radiation, while the PV planning scheme of Scenario 1 is deterministic. Besides, the stochastic method is compared with the robust method. Based on the Monte Carlo simulation, the KNITRO tool plans PVs for Scenario 4 on the MATLAB platform [50], [51]. In Table II, the bolded solar radiation represents the original design parameters of the corresponding solution, while the remaining unbolded parameters represent the performance of the solution in extreme environments such as when $u = 300$. As shown in Table II, robust planning has better effectiveness than deterministic planning under extreme solar radiation. Compared with the stochastic method, although the robust method is more conservative in terms of operational benefits under a normal state, it has higher performance in an extreme environment without accurate knowledge of the PDF. Especially, for the extreme environment ($u = 300$), the objective function of robust planning in scenario 4 is reduced by 12.3% compared to stochastic planning in scenario 4, and is reduced by 14.6% compared to deterministic planning in scenario 1. Robust planning can improve the stability and reliability of the grid connection, optimize power generation, and reduce maintenance costs. Therefore, the proposed method is suitable for scenarios with irregular solar radiation or high operational stability requirements, which has higher robustness than the current stochastic planning and deterministic planning schemes mostly used in SDN.

Considering Scenario 2 as an example, the robust planning results of different adjustable margin Γ values are shown in Table III. When $\Gamma = 0$, the solar radiation is the desired solar radiation value, indicating that the randomness of solar radiation is not considered and the solar radiation at each node is regarded as a constant value. In this case, the adjustable robust optimization planning will be transformed into the traditional PV deterministic planning. When $\Gamma < 68$, the predicted value of solar radiation at some nodes is considered accurate, while the uncertainty of solar radiation at only a portion of the nodes is

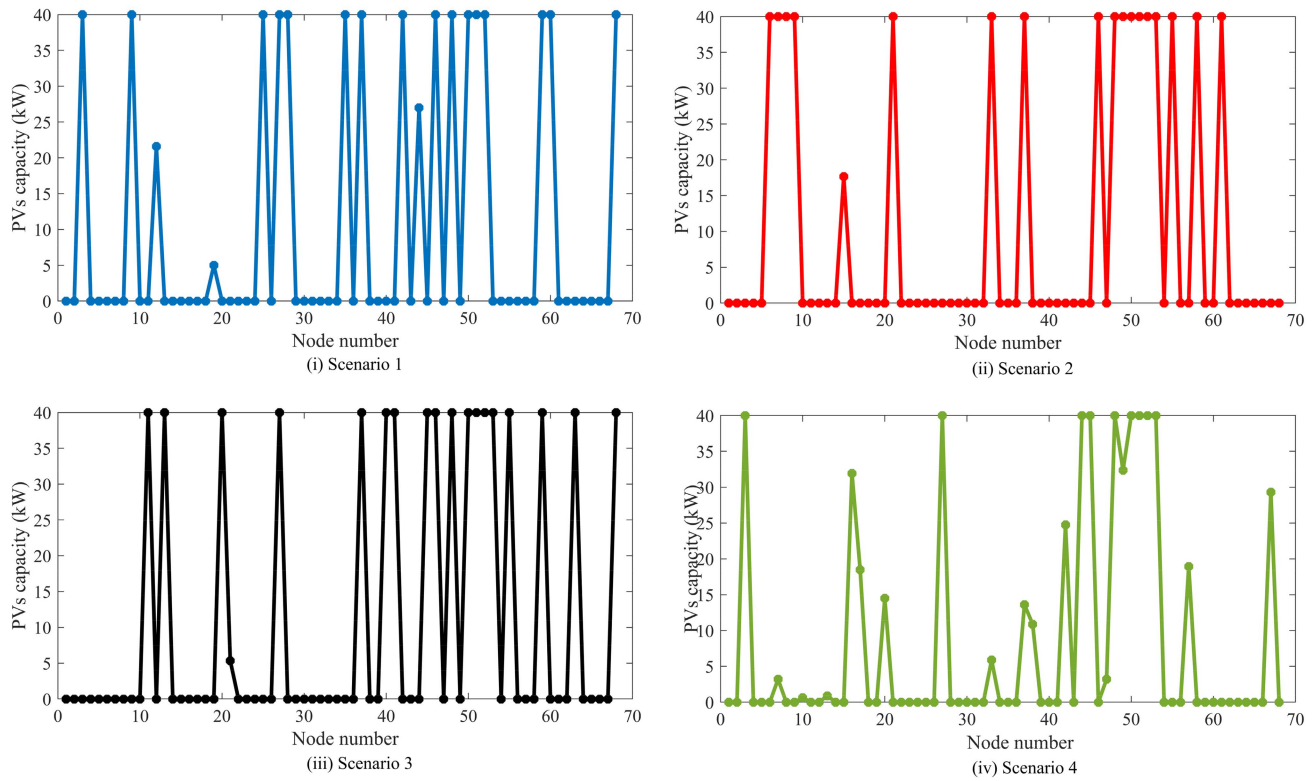


Fig. 7. Installation capacity at each scenario.

TABLE III
ADJUSTABLE ROBUST PLANNING

Adjustable margin	Objective function value (\$)
$\Gamma=68$	1106009
$\Gamma=48$	1082312
$\Gamma=28$	1061162
$\Gamma=18$	1052882
$\Gamma=8$	1046921

considered. As the adjustable margin Γ increases, the extreme scenario of the system gradually worsens, which leads to more uncertainty factors. When the solar radiation of some nodes is low, the actual PV generation is insufficient and the nearby loads are not sufficiently compensated by the PV, thus making the active power loss in the distribution network increase. Therefore, the operational efficiency of the distribution network will deteriorate, and the objective function value will increase. Γ allows the conservativeness of the PV planning to be adjusted, and managers can balance the trade-off between investment cost and robustness accordingly. During the actual testing, it can be found that the program preferred nodes 49-51 as uncertain nodes when the adjustable margin is small, indicating that nodes 49-51 have a great effect on the deterioration of the objective function. Combined with the analysis in Fig. 7 and Table II, setting PV at nodes 49-51 avoids the most inferiority of the objective function when the solar radiation is small, which helps to improve the robustness performance of the SDN and reduce SDN operating

costs. However, stochastic planning and deterministic planning neglect the most inferiority of the objective function when the solar radiation is insufficient, which reduces the robustness performance of the system.

C. Optimization Method Comparison

For the proposed PV robust planning model, the optimization effectiveness of GWO-IPSO, PSO, Atomic Orbital Search (AOS) [51], Taboo Search (TS) [52], and GWO methods are compared in Fig. 8. The following conclusions can be drawn:

- 1) In terms of optimization effectiveness, the final optimization effectiveness of traditional PSO is much lower than that of AOS, GWO, and GWO-IPSO. The effectiveness of GWO-IPSO is 12.19%, 15.78%, and 56% higher than that of GWO, AOS, and TS. The integration of the local search capabilities of GWO and the global search capabilities of IPSO has resulted in excellent optimization capabilities. The proposed GWO-IPSO method improves convergence and the ability to keep out of the local optimum.
- 2) In terms of the calculation and convergence speed, the computation time of 5 minutes in this scenario can add up to weeks of delay in a larger-scale SDN. GWO has the fastest calculation speed, requiring 97.24s for the objective function to reach a steady state. Compared with the other methods, GWO exhibits stronger local searchability but weaker global searchability. The calculation speed of the proposed GWO-IPSO method is 51% lower than that of PSO, but the convergence speed and the optimization

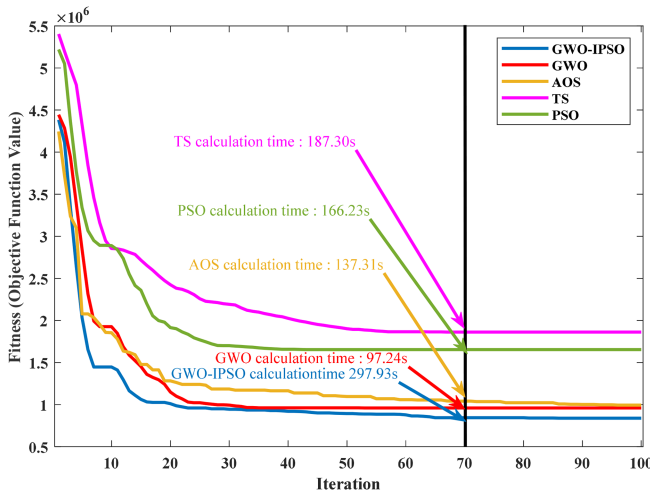


Fig. 8. Optimization algorithm convergence comparison for scenario 4.

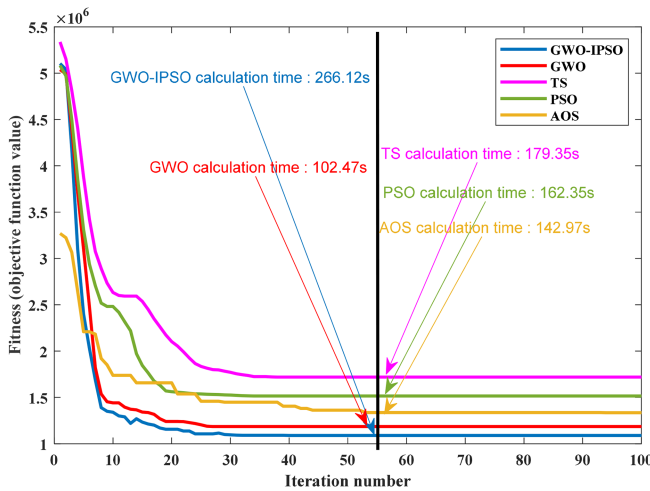


Fig. 9. Optimization algorithm convergence comparison in Scenario 3.

effect are increased by 29.2% and 52.9%. While saving time in the calculation may increase efficiency, ignoring robustness and reliability may lead to an infeasible or unstable plan, which has adverse effects on the construction and operation of PV power plants in the SDN.

To further compare the performance of different optimization methods, the comparison of optimization methods in Scenario 3 is shown in Fig. 9. Similar to Scenario 4, the optimization effect of our proposed GWO-IPSO method and PSO is much better than that of GWO. Compared with other optimization methods, the proposed GWO-IPSO method improves accuracy at the cost of acceptable calculation time.

VI. CONCLUSION

This paper proposed a PV adjustable robust planning model based on a novel GWO-IPSO integration, which optimizes PV sizing and PV positioning considering solar radiation uncertainty. The adjustable robust optimization model quantifies the variable uncertainty in a set, where the economic objective function and constraints are transformed into a robust bi-level

optimization problem using the linearization theory. Then, the bi-level optimization problem is transformed into an equivalent single-layer optimization problem by the strong duality theory. Eventually, the proposed GWO-IPSO method is used to deal with the robust optimization problem. The practicability of the proposed method is demonstrated based on the IEEE 69-bus radial distribution network. For distributed PV planning with uncertainty in solar radiation, the proposed method can realize the adjustable PV configuration in extreme scenarios and increase the robustness of the optimization results. Compared with stochastic optimization methods, the adjustable robust optimization model reduces the requirements for probabilistic data. Compared with the traditional AOS, TS, GWO, and PSO methods, the proposed GWO-IPSO method delivers better convergence and searchability, which can also be applied to a broad range of application scenarios by combining other physical models. In the future, the presented model will consider the stochastic feature of other meteorological data and analyze its impact on the planning results.

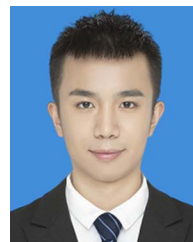
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